



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering
MAT 213: Statistics and Probability
Mid-Term Examination
Spring 2024

24th Batch
29 June 2024

3rd Semester
Time: 1 Hour 30 Minutes

Answer any 3 (three) from the following sets of questions

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

- | Sl. No. | Questions | CLOs | Marks | | | | | | | | | | | | | | |
|---------------|--|---------------|-----------|-------|-------|---------|-------|-------|-----------|---|---|----|----|---|---|--|--|
| 1. | (a) Define statistics. | CLO1 | [02] | | | | | | | | | | | | | | |
| | (b) Write the application of statistics engineering. | CLO2 | [05] | | | | | | | | | | | | | | |
| | (c) Describe and classify data analysis. | CLO2 | [03] | | | | | | | | | | | | | | |
| 2. | (a) Define tabulation and frequency distribution. | CLO1 | [02] | | | | | | | | | | | | | | |
| | (b) Construct a frequency distribution (ungroup data) from the following : | CLO3 | [05] | | | | | | | | | | | | | | |
| | <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">25</div> <div style="width: 50%;">28</div> <div style="width: 50%;">29</div> <div style="width: 50%;">30</div> <div style="width: 50%;">31</div> <div style="width: 50%;">32</div> <div style="width: 50%;">32</div> <div style="width: 50%;">32</div> <div style="width: 50%;">33</div> <div style="width: 50%;">33</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">33</div> <div style="width: 50%;">34</div> <div style="width: 50%;">35</div> <div style="width: 50%;">38</div> <div style="width: 50%;">44</div> <div style="width: 50%;">54</div> <div style="width: 50%;">38</div> <div style="width: 50%;">44</div> <div style="width: 50%;">52</div> <div style="width: 50%;">38</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">44</div> <div style="width: 50%;">51</div> <div style="width: 50%;">37</div> <div style="width: 50%;">43</div> <div style="width: 50%;">50</div> <div style="width: 50%;">37</div> <div style="width: 50%;">43</div> <div style="width: 50%;">47</div> <div style="width: 50%;">37</div> <div style="width: 50%;">43</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">46</div> <div style="width: 50%;">37</div> <div style="width: 50%;">42</div> <div style="width: 50%;">46</div> <div style="width: 50%;">36</div> <div style="width: 50%;">42</div> <div style="width: 50%;">46</div> <div style="width: 50%;">45</div> <div style="width: 50%;">41</div> <div style="width: 50%;">50</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">43</div> <div style="width: 50%;">46</div> <div style="width: 50%;">42</div> <div style="width: 50%;">39</div> <div style="width: 50%;">50</div> <div style="width: 50%;">40</div> <div style="width: 50%;">46</div> <div style="width: 50%;">52</div> <div style="width: 50%;">36</div> <div style="width: 50%;">38</div> </div> | | | | | | | | | | | | | | | | |
| | (c) The frequency distribution of the health condition of 70 workers of a factory as follows: | CLO3 | [03] | | | | | | | | | | | | | | |
| | <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th>Category</th> <th>Frequency</th> </tr> </thead> <tbody> <tr> <td>Good</td> <td>20</td> </tr> <tr> <td>Average</td> <td>37</td> </tr> <tr> <td>Poor</td> <td>23</td> </tr> </tbody> </table> | Category | Frequency | Good | 20 | Average | 37 | Poor | 23 | | | | | | | | |
| Category | Frequency | | | | | | | | | | | | | | | | |
| Good | 20 | | | | | | | | | | | | | | | | |
| Average | 37 | | | | | | | | | | | | | | | | |
| Poor | 23 | | | | | | | | | | | | | | | | |
| 3. | (a) Define presentation of date and central tendency. | CLO1 | [02] | | | | | | | | | | | | | | |
| | (b) Construct a frequency distribution (group data) from the following : | CLO3 | [05] | | | | | | | | | | | | | | |
| | <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">93</div> <div style="width: 50%;">66</div> <div style="width: 50%;">93</div> <div style="width: 50%;">69</div> <div style="width: 50%;">88</div> <div style="width: 50%;">76</div> <div style="width: 50%;">50</div> <div style="width: 50%;">75</div> <div style="width: 50%;">86</div> <div style="width: 50%;">97</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">51</div> <div style="width: 50%;">74</div> <div style="width: 50%;">68</div> <div style="width: 50%;">65</div> <div style="width: 50%;">89</div> <div style="width: 50%;">88</div> <div style="width: 50%;">77</div> <div style="width: 50%;">87</div> <div style="width: 50%;">85</div> <div style="width: 50%;">84</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">82</div> <div style="width: 50%;">83</div> <div style="width: 50%;">82</div> <div style="width: 50%;">81</div> <div style="width: 50%;">79</div> <div style="width: 50%;">80</div> <div style="width: 50%;">65</div> <div style="width: 50%;">74</div> <div style="width: 50%;">69</div> <div style="width: 50%;">54</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">56</div> <div style="width: 50%;">73</div> <div style="width: 50%;">75</div> <div style="width: 50%;">74</div> <div style="width: 50%;">72</div> <div style="width: 50%;">72</div> <div style="width: 50%;">70</div> <div style="width: 50%;">63</div> <div style="width: 50%;">70</div> <div style="width: 50%;">68</div> </div> <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;">59</div> <div style="width: 50%;">67</div> <div style="width: 50%;">61</div> <div style="width: 50%;">60</div> <div style="width: 50%;">56</div> <div style="width: 50%;">62</div> <div style="width: 50%;">72</div> <div style="width: 50%;">73</div> <div style="width: 50%;">71</div> <div style="width: 50%;">57</div> </div> | | | | | | | | | | | | | | | | |
| | (c) The data give the number of fish caught day over a period of 11 days by an angler. Give the five number summary of the following data: | CLO3 | [03] | | | | | | | | | | | | | | |
| | 0 2 5 0 4 4 8 9 8 8 | | | | | | | | | | | | | | | | |
| 4. | (a) What is dispersion and range? | CLO1 | [02] | | | | | | | | | | | | | | |
| | (b) The following table are given the age distribution of 50 employees of a factory: | CLO3 | [05] | | | | | | | | | | | | | | |
| | <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th>Age(in years)</th> <th>25-30</th> <th>30-35</th> <th>35-40</th> <th>40-45</th> <th>45-50</th> <th>50-55</th> </tr> </thead> <tbody> <tr> <td>Frequency</td> <td>3</td> <td>9</td> <td>15</td> <td>12</td> <td>7</td> <td>4</td> </tr> </tbody> </table> | Age(in years) | 25-30 | 30-35 | 35-40 | 40-45 | 45-50 | 50-55 | Frequency | 3 | 9 | 15 | 12 | 7 | 4 | | |
| Age(in years) | 25-30 | 30-35 | 35-40 | 40-45 | 45-50 | 50-55 | | | | | | | | | | | |
| Frequency | 3 | 9 | 15 | 12 | 7 | 4 | | | | | | | | | | | |
| | (i) Find the arithmetic mean. | | | | | | | | | | | | | | | | |
| | (ii) Find median from the above table. | | | | | | | | | | | | | | | | |
| | (iii) Find the mode from the above table. | | | | | | | | | | | | | | | | |

- (iv) Draw a histogram and find the total area of the histogram from the above table.

- (c) Given below are the 1991 census population of Chittagram and Dhaka divisions by sex. Construct a component bar diagram:

CLO3 [03]

Religion	Population		Percentage of population	
	Male	Female	Male	Female
Chittagram	11228	10637	51.3	48.7
Dhaka	17634	16030	52.0	48.0

- (a) Define mean, median and standard deviation.

CLO1 [02]

- (b) The following table contains the population from a village based age group.

CLO3 [05]

Age	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	2	5	7	13	21	16	8	3

- (i) Find mean deviation from mean ($\bar{x}$).

- (ii) Calculate the variance.

- (iii) Calculate standard deviation.

- (c) The ten persons of varying ages weighted and the following weight in kg were recorded.

CLO3 [03]

125 110 117 125 125 136 157 124 110 147

Compute the mean deviation about mean and median and arbitrary value 122.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 213: Digital Electronics

Batch: 24th
Date: 12/06/2024

Mid-Term Examination

Semester: Spring 2024
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer any five questions:

5 × 6 = 30

	CLOs	Marks
1. (a) Explain the digital and analog system.	CLO 1	3
(b) What are the disadvantages of digital techniques?	CLO 1	3
2. (a) Construct a flowchart for repeated-division method of decimal-to-binary conversion of integers.	CLO 1	3
(b) Describe BCD and ASCII codes.	CLO 1	
3. (a) Prove the universality of NAND and NOR gates.	CLO 3	3
(b) What are sum of product and product of sum? If $X = \overline{ABC}(A + D)$, then draw the circuit diagram with a truth table.	CLO 3	3
4. (a) Simplify the expression $Z = ABC + \overline{AB}(\overline{AC})$ and draw circuits for both expressions.	CLO 3	3
(b) Draw a circuit diagram for the equation $X = \overline{ABC} + \overline{DBC} + (\overline{ABD} + \overline{ABC})$.	CLO 3	
5. (a) What is Flip-Flop? Explain the process of setting and resetting NAND Latch.	CLO 4	3
(b) Draw and explain the operation of S-R Flip-Flop.	CLO 4	3
6. (a) Show the Commutative law, Associative Law and Distributive Law in Multivariable Boolean Theorems.	CLO 3	3
(b) Explain DeMorgan's Theorems with example.	CLO 3	
7. (a) Show the step-by-step design procedure for any logic problems.	CLO 3	3
(b) Draw and explain Circuits for Parity Generator and Parity Checker for Even Parity System.	CLO 1	3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE215: Data Structures

Batch: 24th

Date: 10/06/2024

Mid-Term Examination

Semester: Spring 2024

Time: 1 Hour 30 Minutes

Total Marks: 30

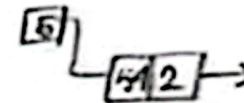
Answer any five questions of the following.

5 × 6 = 30

	CLOs	Marks											
1. (a) Why do we need data structures?	CLO1	1											
(b) Differentiate between linear and non-linear data structures.	CLO1	2											
(c) Explain the operations of data structure.	CLO1	3											
2. (a) Explain the procedure to calculate the length of a one-dimensional array with an example.	CLO1	2											
(b) How a two-dimensional array is represented in the memory?	CLO1	2											
(c) Convert the following two-dimensional array into a one-dimensional array where data are represented in column-major order:	CLO5	2											
$\begin{bmatrix} 5 & 2 & 7 \\ 8 & 6 & 1 \\ 11 & 14 & 4 \end{bmatrix}$													
3. (a) Apply Binary Search algorithm to find the index of 45 in the following array: BIN[11]: <table><tr><td>25</td><td>27</td><td>29</td><td>31</td><td>33</td><td>34</td><td>36</td><td>39</td><td>40</td><td>45</td><td>56</td></tr></table>	25	27	29	31	33	34	36	39	40	45	56	CLO2	3
25	27	29	31	33	34	36	39	40	45	56			
(b) State Bubble Sort algorithm.	CLO2	3											
4. (a) Define linked list. Determine the advantages of a linked list over array.	CLO1	3											
(b) Construct an algorithm to search a Value in a linked list.	CLO4	3											
5. (a) Identify the drawbacks of Binary Search algorithm. Compare between Linear Search and Binary Search algorithms.	CLO4	3											
(b) Suppose, P and T are strings with lengths R and S respectively, and stored as arrays with one character per element. Now, write an algorithm to find the INDEX of P in T.	CLO5	3											
6. (a) Define stack. What are the applications of stack?	CLO1	2											
(b) Consider the following stack with 6 memory cells: CSE: AA, DD, KK, PP, __, __ Describe the stack for the following operations: i. PUSH DD ii. POP iii. PUSH KB iv. PUSH TT	CLO5	4											

7. (a) Suppose, INFO and LNK represent the Information field and Next Pointer field of a linked list respectively. Compose the linked from the given arrays if index of the first node is 5.

	INFO	LNK
0	49	4
1		
2	47	6
3		
4	23	x
⑤	54	2
6	31	8
7		
8	71	0



- (b) Illustrate two-way Linked List. How can you convert a two-way Linked List into a circular list?

CLO1
&
CLO5



HAMMARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT 217
Mid Term Examination
Date: 6/29/2024

Course Title: Ordinary and Partial Differential Equations
Semester: Spring 2024
Total Marks: 30

Answer any Three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

	CLOS	Marks
1. (a) What do you mean by a differential equation?	CLO 1	[1]
(b) Find the general solution of the differential equation $(3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$	CLO 2	[4]
(c) Solve the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$	CLO 3	[5]
2. (a) What do you mean by the order and degree of a differential equation?	CLO 1	[2]
(b) Find the order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 - \frac{dy}{dx} - 2y = e^{-2x}$	CLO 2	[2]
(c) Solve the differential equation $\frac{dy}{dx} + y = xy^3$	CLO 3	[6]
3. (a) What do you mean by an initial value problem?	CLO 1	[1]
(b) Find the solution of the initial value problem $\frac{dy}{dx} = \frac{2y}{x}$ considering $y(0) = 1$	CLO 2	[4]
(c) Solve the differential equation $\frac{d^2y}{dx^2} + \frac{y}{x} = x + e^x$	CLO 3	[5]
4. (a) What do you mean by trajectories of a differential equation?	CLO 1	[1]
(b) Solve the differential equations (i) $\frac{d^2y}{dx^2} + 9y = 0$ (ii) $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$	CLO 2	[6]
(c) Find the general solution of the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$	CLO 3	[3]
5. (a) Write the general form of non-homogenous linear constant co-efficient ODE.	CLO 1	[1]
(b) Find the general solution of the differential equation $\frac{d^2y}{dx^2} + y = \cos 2x$	CLO 2	[4]
(c) Solve the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{2x} \sin x$	CLO 3	[5]

FACULTY OF SCIENCE, ENGINEERING AND TECHNOLOGY

ACT 215: Financial and Managerial Accounting

Semester: Spring 2024

24/06/2024

Midterm Examination

Time: 90 minutes

Total Marks: 10x3=30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.
- Answer any three (3) of the following questions.

1 On May 1, 2024 Tony Biden established Biden Travel Agency. The following transactions were completed during the month:

May 1 Invested Tk.85, 500 cash to start the agency.

May 5 Paid Tk.8, 400 cash for office rent.

May 7 Purchased office equipment for Tk. 32,500 cash.

May 10 Incurred Tk.3,700 of advertising costs in the Chicago Tribune, on account

May 15 Paid Tk.3,800 cash for office supplies.

May 17 Withdrew Tk.2,700 cash for personal use

May 20 Earned Tk.37, 000 for services rendered: Tk.24,000 cash is received from Customers and the balance of Tk.13,000 is billed to customers on account.

May 25 Paid Chicago Tribune amount due in Transaction May 10.

May 28 Paid Tk.3,400 for utilities, Tk.4,700 for salaries, Tk.1,450 for rent.

May 30 Received Tk.4, 000 in cash from customers who have previously been billed on May 20.

Requirement: i) Construct a tabular analysis of above transaction. The column headings should be as follows: Cash, A/R, Supplies, Office equipment, A/P, Biden's capital.

ii) Develop an Income Statement and Balance Sheet for the month of May, 2024.

2 Dream Land Park was started on June 1 by Raida. The following selected events and transactions occurred during June.

June. 1 Raida invested Tk.1,70,400 cash in the business.

June. 4 Purchased land costing Tk.95,000 for cash.

June. 8 Incurred advertising expense of Tk.14,800 on account.

June. 11 Paid salaries to employees Tk.2,500.

June. 12 Hired park manager at a salary of Tk.34,000 per month, effective July 1.

June. 13 Paid Tk.3,600 cash for a one-year insurance policy.

June. 17 Withdrew 31,000 cash for personal use.

June. 20 Received Tk.7,700 in cash for admission fees.

June. 25 Sold 400 coupon books for Tk.35.50 each. Each book contains 10 coupons that entitle the holder to one admission to the park.

June. 30 Received \$8,900 in cash admission fees.

June. 30 Paid \$900 on balance owed for advertising incurred on June 8.

Construct Journal for each transaction.

08150

52850

21650

59650

08

02

10

CLO2

CLO3

- 3 a) Define accounting as per AICPA.
b) Identify the users of accounting information and how they use this information.
c) Discuss the different function of accounting
- 4 a) Explain the different objectives of management accounting.
b) Show the different scope of management accounting.
c) Analyze the tools and technique of management accounting
- 5 Akarin Golf and Driving Range was opened on July 1 by Tony Greg. The following selected events and transactions occurred during July:
July 1; Invested Tk.2,20,000 cash in the business.
July 3; Purchased Lee's Golf Land for Tk.78,000 cash. The price consists of land Tk.53,000, Buildings Tk.15,000 and equipment Tk.10,000
July 5; Advertised the opening of the driving range and miniature golf course, paying advertising expenses of Tk.3,600.
July 6; Paid Cash Tk.1,480 for a one-year insurance policy.
July 10; Purchased golf clubs and other equipment for Tk.2,600 from Parton company payable in 30 days.
July 18; Received Tk.800 in cash for golf fees earned.
July 19; Sold 100 Coupon books for Tk.15 each.
July 25; Withdrew Tk.1000 cash for personal use.
July 30; Paid Parton Company in full.
July 30; Received Tk.1000 cash for fees earned.
Construct Journal for each transaction.

02
04
04

CLO1

03
04
04

CLO1

10

CLO3



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering (CSE)
Faculty of Science, Engineering and Technology

Code: MAT 217
Final Examination
Date: 10/7/2024

Total Time: 2.00 Hours

Course Title: Mathematics-III
Semester: Spring 2024
Total Marks: 40

Answer any four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

		CLOs	Marks
Q1.	(a) What do you mean by a partial differential equation?	CLO1	[1]
	(b) Show the differential equation of all spheres of radius λ , having center in the x-y plane is $z^2 \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1 = \lambda^2$	CLO2	[4]
	(c) Form a partial differential equation by eliminating arbitrary constants from i. $z = axe^y + \frac{a^2 e^{2y}}{2} + b$ ii. $z = ax + a^2 y^2 + b$	CLO3	[5]
Q2.	(a) Find the order and degree of the partial differential equation $\frac{\partial z}{\partial x} - c \frac{\partial^2 z}{\partial y^2} = 0$.	CLO1	[1]
	(b) Show that the partial differential equation form by eliminating the arbitrary function from $\varphi(x^2 + y^2 + z^2, x^3 - y^3 - z^3) = 0$ is $(yz^2 - y^2z)p - (zx^2 + xz^2)q = xy^2 + x^2y$. Where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.	CLO2	[5]
	(c) Form the partial differential equation by eliminating the arbitrary function from $\varphi(x + y + z, x^2 + y^2 + z^2) = 0$	CLO3	[4]
Q3.	(a) What do you mean by a linear and nonlinear partial differential equation?	CLO1	[2]
	(b) Show that the solution of the partial differential equation $2p + 3q = 1$ is in the form $\varphi(3x - 2y, y - 3z) = 0$ Where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.	CLO2	[4]
	(c) Solve the partial differential equation $xzp + yzq = xy$. Where p and q have their usual meaning.	CLO3	[4]
Q4.	(a) If $f(x, y, z, p, q) = 0$ is a partial differential equation. Then write the charprits auxiliary equation.	CLO1	[1]
	(b) Show that the complete integral for $z^2(p^2 z^2 + q^2) = 1$ is $9a^4(ax + y + b)^2 = (a^2 z^2 + 1)^3$	CLO2	[6]
	(c) Solve the partial differential equation $px + qy + p\sqrt{pq} - z = 0$	CLO3	[3]
Q5.	(a) What do you mean by total differential equation?	CLO1	[1]
	(b) Show that the solution of the total differential equation $(yz + xyz)dx + (zx + xyz)dy + (xy + xyz)dz = 0$ is in the form $\ln(xyz) + x + y + z = c$	CLO2	[5]
	(c) Find the general solution of the partial differential equation $(yz + 2x)dx + (zx - 2z)dy + (xy - 2y)dz = 0$	CLO3	[4]



HAMDARD UNIVERSITY BANGLADESH

- Q6. (a) Identify the nature of the partial differential equation $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$ CLO1 [2]
- (b) Identify the nature of the partial differential equation and solve the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} = 0$ CLO2 [5]
- (c) Solve the partial differential equation $(D^4 - D'^4)z = 0$. CLO3 [3]



HAMDARD UNIVERSITY BANGLADESH

Department of CSE
Faculty of Science, Engineering and Technology
B.SC (Honours) in CSE
BAN211 : History of Emergence of Bangladesh
Final Examination
Spring 2024

3rd Semester
25 September 2024

Time: 02.00 hours

Total Marks: 40

Marks are shown in the right margin

Marks

GROUP A

Answer any two of the three questions

10×2=20

4. Give the ethnic identity of the People of Bangladesh
5. Describe about ancient janapadh of Bangladesh
6. Discuss the origin and evaluation of Bangla language

CLO1

GROUP B

Answer any two of the three questions

10×2=20

4. Explain the significance of Language Movement in 1952
5. Briefly discuss the 'Two nation theory'
6. Discuss the Features and importances of Lahore Resolution

CLO2



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering

Faculty of Science, Engineering and Technology

Semester Final Examination 2024

Course Code: MAT213

Course Title: Statistics and Probability

Final Examination

Semester: Spring 2024

Date: 5/10/2024

Total Time: 2.00 Hours

Total Marks: 40

Answer any four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- | No. | Questions | CLOs | Marks | | | | | | | | | | | | |
|---------------------|---|-----------------|-------|-----|-----|---|---|---------------------|-----|-----|-----|-----|-----|--|--|
| Q1. | (a) Define probability, event and exhaustive event. | CLO1 | [2] | | | | | | | | | | | | |
| | (b) Let X be a discrete random variable with the following probability distribution: | CLO3 | [4] | | | | | | | | | | | | |
| | <table border="1"> <tr> <td>Value of $X: x$</td> <td>-7</td> <td>-4</td> <td>-1</td> <td>2</td> <td>5</td> </tr> <tr> <td>Probability: $f(x)$</td> <td>0.2</td> <td>0.1</td> <td>0.3</td> <td>0.3</td> <td>0.1</td> </tr> </table> | Value of $X: x$ | -7 | -4 | -1 | 2 | 5 | Probability: $f(x)$ | 0.2 | 0.1 | 0.3 | 0.3 | 0.1 | | |
| Value of $X: x$ | -7 | -4 | -1 | 2 | 5 | | | | | | | | | | |
| Probability: $f(x)$ | 0.2 | 0.1 | 0.3 | 0.3 | 0.1 | | | | | | | | | | |
| | (i) Find the probability of $X = -4$ | | | | | | | | | | | | | | |
| | (ii) $P(X \leq -4)$ | | | | | | | | | | | | | | |
| | (iii) $P(X \geq -4)$ | | | | | | | | | | | | | | |
| | (iv) $P(-4 \leq X \leq 5)$ | | | | | | | | | | | | | | |
| | (v) Find the value of X . mean | | | | | | | | | | | | | | |
| | (vi) Variance of X . | | | | | | | | | | | | | | |
| | (c) Let X be a continuous random variable with the probability density function | CLO3 | [4] | | | | | | | | | | | | |
| | $f(x) = \frac{6}{8}x^2(2-x); 0 \leq x \leq 3$. Find | | | | | | | | | | | | | | |
| | (i) $P[X < 2]$ | | | | | | | | | | | | | | |
| | (ii) $P[X > 2]$ | | | | | | | | | | | | | | |
| | (iii) $P[1.5 < X < 2]$ | | | | | | | | | | | | | | |
| | (iv) Mean of X . | | | | | | | | | | | | | | |
| | (v) Variance of X . | | | | | | | | | | | | | | |
| Q2. | (a) Write a short note about random variable. | CLO1 | [2] | | | | | | | | | | | | |
| | (b) If X is a beta variate of first kind with parameter l and m , then prove that the r^{th} moment of this distribution is: | CLO2 | [4] | | | | | | | | | | | | |
| | (i) $\mu'_r = \frac{\Gamma(l+m)\Gamma(l+r)}{\Gamma(l+m+r)\Gamma l}$ | | | | | | | | | | | | | | |
| | (ii) Mean = $\frac{l}{l+m}$ | | | | | | | | | | | | | | |
| | (c) Sharmim and Iqbal company estimates the net profits on a new product; it is launching to be Tk. 3 lac during the first year if it is successful Tk. 1 lac if it is moderately successful and a loss Tk. 1 lac if it is unsuccessful. The company assigns the following probabilities to first year prospects for the product, successful: 0.15, moderately successful: 0.25, and unsuccessful: 0.60. What are the expected value and standard deviation of the first year net profit for the product? | CLO3 | [4] | | | | | | | | | | | | |
| Q3. | (a) Define marginal probability and conditional probability function. | CLO1 | [2] | | | | | | | | | | | | |
| | (b) Let X and Y have the following distribution: | CLO3 | [4] | | | | | | | | | | | | |
| | $f(x, y) = \begin{cases} ky^2, & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$ | | | | | | | | | | | | | | |
| | Determine the constant function k and find $P(X \leq 1)$. | | | | | | | | | | | | | | |
| | (c) Let X and Y have the following distribution: | CLO3 | [4] | | | | | | | | | | | | |



HAMDARD UNIVERSITY BANGLADESH

$$f(x,y) = \begin{cases} 2(x+y-2xy); & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

- Check that $f(x,y)$ is a density function.
- Find the marginal density function.
- Determine the conditional probability function of x given y .

- Q4. (a) Define binomial distribution. CLO1 [2]
(b) The variance and mean of a binomial distribution are $\frac{6}{3}$ and 6 respectively. CLO3 [4]
Find
i. Probability function.
ii. Probability($x = 0$).
iii. Probability($x \geq 1$).
iv. Justify the distribution.
(c) Out of 400 families with 5 children each, how many would you expect to have CLO3 [4]
i. 3 boys 0.3125
ii. 5 girls 175
iii. Either 2 or 3 boys?
Assume equal probabilities for boys and girls.
- Q5. (a) Write a short note about Poisson distribution. CLO1 [2]
(b) Show that for Poisson distribution, Mean = Variance = 1. CLO2 [4]
(c) If 3% of the electric bulbs manufactured by a company are defective. Find the CLO3 [4]
probability that in a sample of 100 bulbs are:
i. 0 defective
ii. 1 defective
iii. 2 defective
iv. At least one defective
v. 5 bulbs will be defective.
- Q6. (a) Define normal distribution. CLO1 [1]
(b) Prove that the mean and variance of the standard normal variate are 0 and 1 respectively. CLO2 [4]
(c) The income of a group of 10000 persons was found to be approach distributed CLO3 [5]
with mean TK. 1750 p.m. and standard deviation TK. 50. Show that of this
group 95% had income exceeding Tk. 1668 and only 5% had income exceeding
TK. 1832. What was the lowest income among the richest 100%?

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 213: Digital Electronics

Batch: 24th

Date: 30/09/2024

Final Examination

Semester: Spring 2024

Time: 2 Hours

Total Marks : 40

Answer any **five** questions:

5 × 8 = 40

- | | | |
|--|------|---|
| 1. (a) Write the working principles of the IC 74ALS174 (Parallel in- Parallel out) with diagram. | CLO1 | 4 |
| (b) Design a Full-adder with truth table. | CLO1 | 4 |
| 2. (a) Sketch a 16-bit synchronous Up/Down counter. | CLO4 | 4 |
| (b) Draw a MOD-29 synchronous Up counter using D flip-flop. | CLO4 | 4 |
| 3. (a) Write the differences between Asynchronous and Synchronous counter. | CLO4 | 4 |
| (b) Explain propagation delay in ripple counter with an example. | CLO4 | 4 |
| 4. (a) Illustrate a Hexa-decimal De-multiplexer. | CLO2 | 4 |
| (b) Draw and explain an eight input digital Multiplexer. | CLO2 | 4 |
| 5. (a) Write the steps of a small project management in digital electronics. | CLO3 | 4 |
| (b) List the commonly used IC packages. | CLO3 | 4 |
| 6. (a) Which memory stores the most bits: a 7M × 16 memory or a memory that stores 2M words at a word size of 32 bits? | CLO4 | 4 |
| (b) Draw and explain a Digital-ramp Analog to Digital Converter. | CLO2 | 4 |
| 7. (a) Differentiate between Static RAM and Dynamic RAM. | CLO4 | 4 |
| (b) Illustrate CPU-Memory connections. | CLO4 | 4 |

FACULTY OF SCIENCE, ENGINEERING AND TECHNOLOGY
ACT 215: Financial and Managerial Accounting

Semester: Spring 2024

02/10/2024

Final Examination 2024

Time: 120 minutes

Total Marks: 10x4=40

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.
- Answer any Four (4) of the following questions.

- 1 Judi Dench is a licensed CPA. During the first month of operations of her business, the following events and transactions occurred: 10

April 1 Judi invested Tk.25,000 cash

April 2 Hired a secretary receptionist at a salary of Tk.1200 per month.

April 3 Purchased Tk.1,700 of supplies on account from Read supply Company.

April 7 Paid office rent of Tk.900 cash for the month.

April 11 Completed a tax assignment and billed client Tk.2,100 for services provided.

April 12 Received Tk.3,500 advance on a management consulting engagement.

April 17 Received cash of Tk.1,200 for services completed for H. Arnold co.

April 31 Paid Secretary receptionist Tk.1200 salary for the month.

April 31 Paid 60% of balance due Read Supply Company.

Requirement: Transactions post into the ledger.

- 2 At the end of first month of operations, Wastson Answering Service has the following unadjusted trial balance: 10

Watson Answering Service
Trial Balance, June 30, 2024

Titles of Accounts	Tk.	Tk.
Cash	7,400	
Accounts Receivable	3,800	
Prepaid Insurance	2,400	
Supplies	1,300	
Equipment	62,000	
Notes payable		40,000
Accounts Payable		2,100
Watson's Capital		35,000
Watson's Drawing	1,000	
Service Revenue		4,900
Salaries Expenses	3,200	
Utilities Expenses	800	
Advertising Expense	400	
	<u>82,300</u>	<u>82,300</u>

Other data:

- Insurance expires at the rate of Tk.500 per month.
- There are Tk. 500 of supplies on hand June 30, 2024.
- Depreciation on the equipment is Tk.900.
- Interest of Tk.500 on the notes payable has accrued during June.

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What do you mean by Trial Balance?

The following Trial Balance of Mr. James Gomez has been prepared incorrectly. You are asked to prepare the Trial Balance correctly:-

Trial Balance
April 30, 2024

Account titles	Debit(TK)	Credit (Tk)
Capital – Mr. Gomez	1,90,000	
Drawings- Mr. Gomez		12,480
Furniture		12,000
Import Duty	2,000	
Export Duty	3,000	
Wages	18,720	
Allowance for doubtful debt	2,000	
Bad Debt	740	
Insurance Premium	240	
Buildings	30,000	
Transport in		32,900
Transportation out	5,000	
Salaries	2,500	
Sales		2,07,700
Purchases		1,42,230
Accounts Payable	12,790	
Notes Receivable	1,500	
Notes payable	18,460	
Sales Return		11,760
Purchase Return		15,000
Accounts Receivable		37,800
Investment		15,000
Cash	40,000	
Bank Loan	12,270	
Advertisement		4,000
Inventory		26,420